Chapter 13

Assessment and anesthesia for thoracic surgery

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Anesthetic considerations

Key facts

- Specific anesthetic considerations in thoracic surgery include the need for accurate assessment of the airway, need for lung isolation and bronchoscopy prior to procedure, increased use of epidural anesthesia, and greater need for invasive monitoring, particularly arterial blood gases and pressures, with location depending on procedure.
- The ***Mallampati classification*** is used to identify patients who may have difficult airways.
- Bronchoscopy is routinely performed after intubation, and may impact on conduct of anesthesia in several ways.
- Tracheal reconstruction has several unique anesthesia issues.

Preoperative assessment

Prior to anesthetic for an operation the global function of a patient is described by assigning one of the ***American Society of Anesthesiologists (ASA) Physician Status Classes***. If the procedure is an emergency, than the letter E is added to the class description (📖 p89, Box 3.1).

Evaluation of the airway

- Factors that determine whether a patient has a difficult airway include:
  - Trauma to the head, neck, or face.
  - Ability to open mouth, micrognathia, obstructive teeth, short 'bull' neck, morbid obesity.
  - Diminished cervical spine mobility.
- The airway can be categorized using the ***Mallampati classification*** which determines the degree at which the base of tongue may obstruct a potential view of the vocal cords, prior to intubation. It is assessed by having the patient sit upright with their head in the neutral position, the mouth maximally opened, and the tongue protruded anteriorly. Class I is the easiest potential intubation.
  - Class I: faucial pillars, soft palate, and uvula are visible.
  - Class II: faucial pillars and soft palate are visible, but the uvula is obscured by the base of the tongue.
  - Class III: only the soft palate is visible.
- Class III airway predicts a difficult intubation.

Hemodynamic monitoring

- Basic hemodynamic monitoring should include frequently cycled BP, pulse oximetry, continuous EKG, and occasionally CVP.
- A radial arterial line should be used in any major thoracic surgery procedure, not only for BP monitoring (particularly in patients with epidurals), but also to measure frequent ABGs, especially in single-lung ventilation.
- The arterial line should be placed in the dependent arm when the patient is in the lateral decubitus position.
- For tracheal surgery (📖 pp750–757) or mediastinoscopy (📖 p662), the arterial line should be placed in the left arm, to avoid inaccurate readings during compression of the innominate artery.

Anesthetic considerations for key procedures

Flexible bronchoscopy

- Most flexible bronchoscopy can be performed with topical analgesia with 2–4% lidocaine sprayed onto the vocal cords.
- Conscious sedation can be used with benzodiazepines and narcotics.
- If performed under general anesthetic, *a >7.5mm ET tube* is needed to allow for comfortable manipulation of the bronchoscope.
- Most common complication of flexible bronchoscopy is *hypoxia* from excessive suction aspiration; therefore 100% oxygen should be used.

Rigid bronchoscopy

- General anesthesia required: hypoxia and hypercapnia may complicate procedure.
- Most rigid scopes have a sidearm to allow positive pressure ventilation.
- While anesthesiologists are accustomed to this mode of ventilation, it has to be suspended while the surgeon is working.
- *Intermittent jet O_2 ventilation* may be used: this entrains oxygenated air via a Venturi effect with the advantage that ventilation is not interrupted. Disadvantages are: (1) an iv anesthetic is required as volatile agents are unpredictable given the high gas flow rates, and (2) the anesthesiologist may not be familiar with this type of ventilation.

Mediastinoscopy

BP measurements in the right arm may become inaccurate or unavailable if the innominate artery is intermittently compressed by the mediastinoscope. Therefore arterial lines should be placed in the left arm and a pulse oximeter on placed on the right hand. If the pulse oximeter loses its tracing, the surgeon should be warned that the innominate artery is being occluded.

Tracheal resection and reconstruction

- Airway with critical stenosis may make induction anesthesia difficult.
- Muscle relaxants should be avoided, as the patient may need spontaneous efforts.
- If the stenosis is too narrow to intubate, the surgeon can core out the stenosis with a rigid bronchoscope, or the patient may be intubated above the stenosis and jet/spontaneous ventilation used to proceed.
- Once the trachea is divided, the surgeon can intermittently intubate the distal trachea with a sterile armored *Tovell tube*, while the ET tube is retracted to the proximal airway.
- Just before completion of the tracheal anastomosis, the ET tube is advanced beyond the repair.

Thymectomy for myasthenia gravis

- Patients with MG have autoimmune antibodies to the acetylcholinesterase receptor (AChR), and may be overly sensitive to non-depolarizing muscle relaxants.
- Anticholinesterase medications should be avoided, and patients may require plasmapheresis preoperatively to clear anti-AChR antibodies.
- Non-depolarizing muscle relaxants should be avoided.
- Patients may require a short duration of mechanical ventilation postoperatively secondary to muscle weakness.

Single-lung ventilation

Key facts

- Many thoracic surgical procedures require the surgical lung to be non-ventilated, with selective ventilation of the non-operative lung, with a ***double lumen endotracheal tube (DLT).***
- A ***bronchial blocker*** is an alternative for lung isolation when placement of a DLT would be too difficult, in a patient that is already intubated with a single lumen ET tube or a tracheostomy.

Oxygenation and ventilation

- Oxygenation may be affected by shunting of pulmonary blood flow through the non-ventilated lung.
- Oxygenation should be carefully monitored with measurements of ABGs.
- Tidal volumes of 10–15mL/kg should be used to achieve desired minute ventilation volumes.
- The respiratory rate can be increased to achieve CO_2 levels appropriate for two-lung ventilation.
- Peak airway pressure should be monitored to avoid hyperinflation of the ventilated lung.

Troubleshooting

- *Isolated lung fails to deflate*: suction secretions from the ET tube, confirm appropriate tube position with a pediatric bronchoscope.
- *Oxygenation is inadequate*: suction secretions from the ET tube, confirm appropriate tube position with a pediatric bronchoscope.
  - Add PEEP to the ventilated lung.
  - Increase the fraction of inspired oxygen (FiO_2) to 100%.
  - Briefly inflate the non-ventilated lung.
  - The surgeon can clamp the pulmonary artery of the non-ventilated lung to limit ventilation–perfusion mismatch.
- *Hypercapnia*: briefly inflate non-ventilated lung.

Double lumen endotracheal tubes

- Double lumen tubes are fitted to either the right main stem bronchus (right-sided DLT) or the left main stem bronchus (left-sided DLT). DLT have separate channels to ventilate both the main stem bronchus and the trachea. The right-sided DLT has a 'Murphy's eye' to allow for ventilation of the right upper lobe (Fig. 13.1a).
- Sizing DLT: 39–41F for a man and 35–37F for a woman.
- Left-sided DLTs are adequate for most lung isolation needs, and are easiest to place.
- Right-sided DLTs are harder to place secondary to the need to align the Murphy's eye with the orifice of the RUL bronchus. However, right-sided DLTs are ideal for left pneumonectomy left main stem bronchoplasty procedures and left lung transplants.

Placement of a DLT

- The DLT is inserted with the point of the distal curve facing anteriorly.
- The stylet is removed when the tube enters the trachea.

- Tube is rotated so the bronchial lumen is towards the intended side.
- The tube is advanced until resistance is felt.
- The tracheal cuff is inflated, and the patient is ventilated, allowing for bilateral chest expansion. If satisfactory, then the tracheal side is clamped and tracheal lumen is opened to air, and the bronchial cuff is inflated and the chest is auscultated. Selective ventilation is confirmed.
- Correct positioning is confirmed with a pediatric bronchoscope, immediately after tube placement, and again after positioning patient.
- For a left-sided DLT, the bronchoscope is inserted through the tracheal lumen. The bronchial cuff should be just past the carina in the left main stem. The right upper lobe orifice is identified.
- For right-sided DLT, the bronchial cuff should be just past the carina in the right main stem. The bronchoscope is then inserted down the bronchial lumen, and the Murphy's eye is confirmed to be positioned at the right upper lobe orifice.

Common problems

- Common errors include advancing the tube too far, not far enough, not aligning the Murphy's eye with the right upper lobe bronchus, overinflating the bronchial cuff causing herniation, displacement of the tube with patient positioning.
- Complications of placement include airway injury, pneumothorax, and esophageal intubation.

Bronchial blocker

- Involves placement of a Fogarty type balloon through a single lumen ET tube to selectively occlude an airway.
- Can be used for lung isolation when placement of a DLT would be too difficult, a patient is already intubated with a single lumen ET tube, or a tracheostomy.
- Arndt wire-guided bronchial blocker (Fig. 13.1c): a Fogarty type balloon is attached to a 5–9F catheter with one 1.4mm inner lumen.
- The catheter has a guidewire that passes through the lumen, which can be coupled to a flexible bronchoscope.
- The bronchoscope, with the wire looped, guides the catheter to the correct position, and the balloon is inflated to occluding one bronchus.

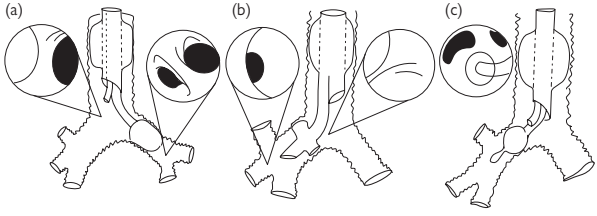


Fig. 13.1 (a) Left-sided, (b) right-sided double-lumen ET tube. (c) Bronchial blocker in right mainstem. Inserts are bronchoscopic views at each level.

Thoracic analgesia

Key facts

- ***Epidural analgesia*** provides excellent analgesia via administration of opioids (e.g., fentanyl) or local anesthetics (e.g., bupivacaine) via a catheter inserted into the epidural space usually before surgery.
- ***Total spinal block*** is a potentially fatal complication of epidural anesthesia.
- ***Intercostal blocks*** provide shorter-term local anesthesia by infiltration of the intercostal nerves above and below a thoracotomy incision: these can be performed at completion of a procedure or postoperatively.

Epidural analgesia

- Thoracic epidural is beneficial in patients undergoing thoracotomy as well as upper abdominal surgery.
- Local anesthetic is continuously applied through a flexible catheter into the epidural space, accessed via a lumbar puncture.
- Anesthetic solution into the pleural space affects the spinal nerve roots by entering the CSF via the dura.
- Epidural infusions include bupivacaine and fentanyl.

Complications

These include complications from placement and side effects of the infusion and inadequate analgesia.

- Placement complications include headache, epidural abscess, and epidural hematoma.
- Infusion side effects include urinary retention, pruritus, hypotension, respiratory depression, and CNS depression.
- *Hypotension:* this is due to vasodilation because of loss of sympathetic vascular tone, and in the case of a complete spinal block (see 'Total spinal block', next bullet point) bradycardia, hypercapnia, and hypoxia. Treat isolated hypotension by:
  - Checking level of block to ascertain it is no higher than dermatome (nipples): if higher reduce infusion rate or discontinue.
  - Giving iv fluid bolus (e.g., 250–500mL normal saline).
  - Start pressor infusion, e.g., 0.03–0.05 micrograms/kg/min norepinephrine (with bolus, e.g., 1mg metaraminol in 10mL normal saline, given in 1mL increments if severe, acute hypotension).
- *Total spinal block*: may occur rapidly and can result in cardiac arrest if not recognized. Usually due to inadvertent intrathecal administration, but can occur with overdose via epidural route causing block to ascend to C8–C4. Warning signs include hypotension, bradycardia, numbness/weakness arms and hands, difficulty breathing, loss of consciousness.
  - ABC: secure airway, ventilated with 100% O_2.
  - Treat hypotension with iv fluids, epinephrine, atropine, pressor.
  - They is no way of reversing a local anaesthetic block: the patient must be ventilated and hemodynamically supported until the local anaesthetic wears off.
  - Naloxone (0.2–0.4 mg iv) reverses effects of opiate analgesia.

- *Epidural headache*: usually postural—worse with standing, relieved by lying down, may be associated with signs of meningism (stiff neck, photophobia, nausea, vomiting, dizziness) and is thought to be due to CSF leak causing traction on the meninges.
  - Small needles and meticulous technique help reduce incidence.
  - Patient can lie down if appropriate.
  - Fluid resuscitation: iv if not tolerating oral fluids.
  - Simple analgesia (acetaminophen, aspirin) and sumatriptan.
  - Caffeine: cola drink or coffee.
  - Blood patch if severe/prolonged: 15–20mL patient's own blood is injected into epidural space to seal CSF leak.
- *Urinary retention*: most thoracic surgery patients will have a Foley catheter in situ, which is discontinued after the epidural.
- *Epidural abscess*: potentially fatal complication. Epidurals are normally discontinued after 3–5 days. Check site daily if kept for longer for inflammation, cellulitis, or pus, particularly if patient develops signs of sepsis postoperatively. If you see these, the anesthesiologist who placed the epidural should be informed, the epidural should be removed, and broad-spectrum iv antibiotics should be started.
- Side effects can be managed by turning the rate of the epidural down or administering naloxone (0.2–0.4mg IV) if the epidural contains an opiate, e.g., fentanyl—naloxone has no ability to reverse the effects of local anesthetic-based epidurals.

Troubleshooting inadequate analgesia

- The anesthesiologist can bolus the epidural to gauge a patient response to determine if it is working.
- The solution concentration, type of analgesia given, and rate may be increased.

Intercostal rib blocks

- Rib blocks are performed at the level of the incision, as well as 2–3 ribs above and below the level of incision.
- Using a 22-G needle, anesthetic agent is administered at the inferior edge of the rib, after an aspiration negative for blood or air.
- Flexible fine-bore catheters can be inserted in the intercostal space at the time of surgery and continuous infusion of local anesthetic (usually bupivacaine) run for several days postoperatively, with the option for bolus top-ups.

Patient-controlled analgesia

- PCA use patient-controlled administration of narcotics such as morphine, fentanyl, hydromorphine, and pethidine (meperidine).
- The computerized system has a loading dose, a timed lockout range in minutes, an hourly limit range in mg, and a potential for a basal rate in mg/h.
- The basal rate option should be avoided in elderly patients or those in an unmonitored unit.

Risk assessment

Key facts

- Performance scores help quantify a patients overall function.
- Major and intermediate cardiovascular risk factors necessitate additional preoperative investigation.
- Postoperative predicted FEV_1 and DLCO are the best indicators for a patient's ability to undergo pulmonary resection.
- The ability to climb a flight of stairs without significant dyspnea is a good surrogate for cardiopulmonary exercise testing.

Major predictors of postoperative complications

- Extent of resection, increased age, pre-existing cardiovascular disease, emergency procedure, immunocompromised patient, current or recent smoker, multiple comorbidities.
- Vital capacity decreases by 25% following a thoracotomy without lung resection, and takes 4–6 weeks to return to preoperative levels.

Performance or Zubrod scores

Performance scores are used to measure and quantify a patient's overall performance status and ability to perform activities of daily living:

- 0: asymptomatic.
- 1: symptomatic but completely ambulatory.
- 2: symptomatic, <50% in bed during the day.
- 3: symptomatic, >50% in bed, but not bedbound.
- 4: bedbound.
- 5: dead.

Assessment of cardiovascular risk

- In patients at risk for coronary artery disease (CAD), a small sub-population of patients with heart failure, exercise-induced arrhythmia, or myocardial ischemia may be identified by non-invasive testing.
- Per 2002/2006 ACC/AHA perioperative clearance guidelines thoracic surgery is an intermediate risk procedure with a 1–5% cardiac risk.
  - In patients with no cardiac history, risk of perioperative MI is 0.07–0.13%.
  - In patients with prior history of MI, risk of a perioperative MI is 2.8–17%.

Initial history and examination

- Includes physical exam, baseline EKG, determine cardiac risk factors by identifying history of CAD, valvular disease, heart failure, arrhythmia.
- Minor risk factors:
  - Diminished activities of daily living such as inability to climb a flight of stairs.
  - Age.
  - Abnormal EKG.
  - AF or other abnormal rhythms that are controlled.
  - Hypertension.
  - History of stroke.

- *Intermediate risk factors*:
  - Canadian class I or II angina.
  - Compensated CHF.
  - Chronic renal insufficiency.
  - Diabetes mellitus.
  - Prior MI.
- *Major risk factors:*
  - MI within 30 days.
  - Unstable or Canadian class III or IV angina.
  - Decompensated CHF.
  - Significant arrhythmias, including AF with rapid ventricular response, AV block.
  - Severe valvular disease.

ACC/AHA guidelines

- Patients with no cardiovascular risk factors do not require additional cardiovascular testing. For patients with minor or intermediate risk factors additional cardiovascular testing may be warranted, on a case-by-case basis.
- Patients with minor risk factors may proceed to surgery without additional cardiovascular testing.
- Patients with intermediate risk factors but good exercise capacity (e.g., can walk up a flight of stairs without dyspnea), can proceed with out additional cardiovascular testing.
- Patients who have undergone surgical or procedural revascularization within 5 years and who are asymptomatic may proceed without additional cardiovascular testing.
- Patients with one or more major risk factors require additional cardiovascular testing.

Cardiovascular investigations

- In patients where significant coronary artery disease is the main concern, non-invasive tests include myocardial perfusion stress/rest imaging (📖 p28), and stress echocardiography (📖 p25). Areas of reversible ischemia require invasive coronary imaging.
- Coronary catheterization identifies significant coronary stenoses, which would need treatment regardless of planned pulmonary resection. For lesions than can be treated with PCI, *bare metal* stents are favoured in this context, because of the potentially catastrophic sequelae of stent thrombosis associated with stopping antiplatelet therapy remains a risk for much longer with drug-eluting stents:
  - Clopidogrel increases bleeding risk: although surgery can be carried out safely, most surgeons prefer to discontinue clopidogrel for 5–10 days preoperatively.
  - It is usually safe to do this 6 weeks after angioplasty or bare metal stent, but probably not for ≥1 year after drug-eluting stent.
  - Planned thoracic surgery should be delayed 6 weeks after CABG.

Pulmonary assessment

Key facts

Pulmonary function testing allows for the assessment of lung volume, air flow, lung mechanics, and gas exchange.

Spirometry

- Measures expiratory airflow volume over time.
- 'Normal' reference values are based on age, gender, and height.

Forced vital capacity and forced expiratory volume in 1s

- FVC and FEV_1 are probably the most important measurements for assessing risk of postoperative pulmonary insufficiency.
- Most commonly used spirometry tests.
- Value after bronchodilator therapy should be used.
- % predicted (predicted value based on age, gender, and height) is better representative of pulmonary function than absolute value.
- Obstructive conditions (see Table 13.1) result in reduced FEV_1, reduced FEV_1/FVC is indicative of obstructive lung disease secondary to a larger total lung capacity from air trapping, and reduced elastic recoil of the lung.
- Restrictive lung disease (see Table 13.1) results in reduced FVC and FEV_1 and possible preserved FEV_1/FVC ratio.
- FEV_1 of 2L, or >80% predicted is suitable for pneumonectomy, and FEV_1 >1.5L is suitable for lobectomy.

Predicted postoperative (ppo) FEV_1 and DLCO

- If either the FEV_1 or DLCO is <80% predicted, postoperative lung function should be predicted through additional calculation or testing.
- Ppo FEV_1 and ppo DLCO values are more accurate in predicting operative risk.
- Ppo FEV_1 and DLCO are calculated by estimating the quantity of lung parenchyma lost after surgical removal.
- For lobectomy, counting the number of anatomic segments removed is used for calculation, whereas lung perfusion measured by quantitative radionuclide perfusion should be taken into account in borderline cases and in pneumonectomy (Box 13.1).

Table 13.1 Pulmonary disorders

Obstructive disorders	Restrictive disorders
Asthma	Sarcoidosis
Chronic bronchitis	Congestive heart failure
Emphysema	Pulmonary fibrosis
Cystic fibrosis	Fibrosing alveolitis
Bronchiolitis obliterans	Thoracic deformities
Eosinophilic granulosa	Interstitial pneumonitis
Lymphangiomyomatosis	

Gas exchange

- Gas exchange tests measure alveolar oxygen exchange.
- Diffusing capacity of the lung for carbon monoxide (DLCO) Identifies patients with restrictive lung disease such as interstitial lung disease and other parenchymal abnormalities.

Predicting postoperative FEV_1

$$\text{ppo } FEV_1 = \text{Pre-op } FEV_1 \times \frac{19 - \text{number of segments to be resected}}{19}$$

$$\text{ppo } FEV_1 = \text{Pre-op } FEV_1 \times (1 - \text{fraction of total perfusion for resected lung})$$

- DLCO has been demonstrated to be a predictor of postoperative morbidity and mortality.
- Low preoperative DLCO correlates with hospital readmission and poorer quality of life indexes.
- On average the right lung contributes 55% of perfusion and the left lung 45%.
- ppo FEV_1 or DLCO <40% indicates an increased risk for perioperative and cardiopulmonary morbidity after pulmonary resection. Patients should undergo additional exercise testing to confirm inoperability.
- A ppo FEV_1 or DLCO <30% may be a better threshold.

Cardiopulmonary exercise testing (CPET)

- CPET records the exercise EKG, heart rate response, minute ventilation, and maximal oxygen consumption (VO_2max).
- A VO_2max of 10–15mL/kg/min indicates an increased risk of morality after pulmonary resection.
- For patients with a VO_2max of <10mL/kg/min or <15mL/kg/min and a ppo FEV_1 and/or DLCO <40%, non-operative therapy is recommended.[1]
- Stair climb of 1 flight (4m) is a good surrogate.
- Inability to climb a flight of stairs without dyspnea corresponds to a VO_2max of <10mL/kg/min.[2]
- Combined lung volume reduction surgery (LVRS) and cancer resection.
- LVRS has shown to be effective in patients with heterogenous emphysema and upper lobe predominance and poor exercise capacity (📖 p726).
- For patients with poor lung function, upper lobe cancer, and upper lobe predominant emphysema, combined LVRS and lung cancer resection should be considered if the FEV_1 and/or DLCO are >20% predicted.[1]

References

1. Colice GL *et al.* Physiologic evaluation of the patient with lung cancer being considered for resectional surgery. *Chest* 2007; **132**:161S–77S.
2. Pollock M *et al.* Estimation of ventilatory reserve by stair climbing. A study in patients with chronic airflow obstruction. *Chest* 1993; **104**(5):1378–83.

Flexible bronchoscopy

Key facts

- Flexible bronchoscopy is the evaluation of the bronchial airways, for diagnostic or therapeutic purposes, using a flexible bronchoscope equipped with fiberoptic bundles, or with new-generation scopes, a charge coupled device at the end of the scope for improved optics.
- Flexible bronchoscopy can be readily accomplished with topical analgesia (awake patient) and therefore allows assessment of vocal cords and laryngeal function, and tracheomalacia (📖 p750), in addition to tracheobronchial toilet and biopsy.
- The most common complication of bronchoscopy is hypoxia from excessive suction aspiration.

See Fig. 13.2 for bronchial anatomy.

Procedure

- Flexible bronchoscopy can be performed via general anesthesia (7.5 ET tube or larger for standard adult scope), conscious sedation, or topical analgesia.
- The scope can be introduced through the nose, mouth, ET, or tracheostomy tube or stoma, or a rigid bronchoscope and can visualize the airway down to subsegmental bronchi.
- If you are doing this for the first time, familiarize yourself with the flexion and extension position, and the up, down, and neutral positions.

Setup

- There should be a nurse or other clinical assistant present.
- You need a clean bronchoscope, image processor, and light source (bronchoscopy stack), aerosolized local anesthetic for awake patients and adequate sedation on board for intubated patients (for whom you also need the ET tube adaptor so they can still be ventilated during the procedure), 1L sterile normal saline, and a 10mL syringe with a straight nozzle for flushing, suction and suction tubing, specimen trap, flexible biopsy forceps, needle, and specimen pots.
- In case of emergencies have readily available equipment to perform bag and mask ventilation, and 1:10 000 epinephrine to control endobronchial bleeding.
- In awake bronchoscopy the upper airway is anesthetized by spraying 1–2% lidocaine aerosol into the nostril and onto posterior oropharynx. A bite block should be used if the oral route is planned.

Airway inspection

- Introduce scope and visualize vocal cords: in awake bronchoscopy spray lidocaine via bronchoscope which will induce coughing but eventually suppress the cough reflex.
- Instruct patient to take a deep breath and pass scope through open cords in trachea.
- Maintain orientation at all times: the soft, flat membrane is posterior.
- Systematically inspect the trachea, mainstem bronchi and segmental bronchi for size, patency, erythema, thrombi, secretions, masses.
- It is vital to use a system to ensure that each segment is correctly identified and none are missed.

Diagnostic applications

- *Brushing*: brush enclosed within a flexible catheter is advanced out of the sheath once near the lesion, which is then briskly rubbed a few times, retracted, and washed in saline/smeared on a microscopy slide.
- *Bronchoalveolar lavage*: a specimen trap is attached to the suction port on the handle of the scope which is advanced into the subsegmental bronchi, 10–20mL of sterile saline flushed down the scope and then aspirated into the trap, and sent for cytology. The difference between this and bronchial washings, is that the scope is wedged in a small airway in BAL to increase the specificity of the sample to the distal airway (washings are contaminated with oral flora).
- *Endobronchial biopsy*: flexible biopsy forceps advanced onto lesion under bronchoscopic vision, closed, and pulled quickly back.
- *Transbronchial biopsy*: this is a method of blindly sampling lung parenchyma using biopsy forceps. The patient is instructed to take a deep breath, the forceps are opened at the limit of the airway, the patient exhales, the forceps are closed, and if the patient does not complain of pain (indicating that the visceral pleura has been breached, risking pneumothorax) the forceps are tugged back.

Therapeutic applications

Aspiration of secretions, laser treatment, photodynamic therapy (PDT), argon plasma coagulation (APC), cryotherapy, stent placement, foreign body retrieval, and Wang needle transbronchial needle aspiration (TBNA).

Complications

- The most common complication during flexible bronchoscopy is hypoxia from suction aspiration.
- Other complications are rare, and include bleeding, pneumothorax, and respiratory failure (~0.1–0.2% each)
- The incidence of pneumothorax is ~4% in transbronchial biopsy.
- Self-limiting fever is not uncommon after BAL.

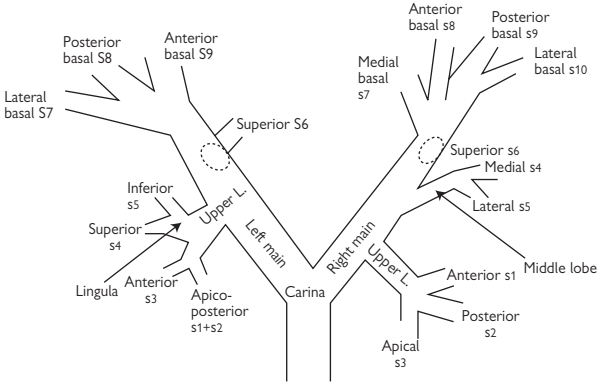


Fig. 13.2 Bronchoscopic anatomy.

Rigid bronchoscopy

Key facts

- Rigid bronchoscopy involves a stainless steel tube equipped with a removable lens that can allow for direct visualization, or insertion of a Hopkins's telescope for magnified camera visualization.
- Rigid bronchoscopy is best performed with general anesthesia.
- The rigid bronchoscope is useful for diagnostic and therapeutic interventions. As a flexible bronchoscope can be inserted down a rigid bronchoscope, all the therapeutic and diagnostic manoeuvres of a flexible bronchoscope can be performed through a rigid bronchoscope.
- Endobronchial ultrasound transbronchial needle aspiration (EBUS-TBNA) is an adjunct to mediastinoscopy, although cervical mediastinoscopy is the gold standard for evaluation of nodal tissue and other pathology of the superior and middle mediastinum.

Procedure

Set-up

- It is generally performed under general anesthesia, and it is inserted under direct visualization through the vocal cords (Fig. 13.3).
- The patient can be ventilated via intermittent insufflations through a side port, or Venturi (jet) ventilation (📖 p159).
- Sedate and preoxygenate the patient using bag and mask ventilation.
- Protect patient's eyes, lips, and teeth with eye pads and mouth guard.
- Place the patient's head in the sniffing position, standard for intubation.

Rigid bronchoscopy intubation

- Insert scope into mouth, advance under direct vision posterior to the tongue.
- The upper row of teeth is protected by the examiner's thumb at all times: resist any temptation to lever the scope against the teeth.
- By elevating the tongue the epiglottis is identified (Fig. 13.3).
- Raise the tip of the epiglottis anteriorly with the bronchoscope to visualize the posterior part of the vocal cords. Alternatively a laryngoscope can be used.
- Advance the bronchoscope towards the visual cords and turn the bronchoscope 90° to align the vertical orifice between the cords with the bronchoscope.
- Gently advance the scope through the cords into the trachea.
- The rigid bronchoscope can also be used to core out tumours or granulation tissue, or a flexible scope can be introduced (📖 p654).

Complications

- Injury to oropharyngeal structures.
- Hypoxia.
- Hypercapnia.
- Airway injury.
- Pneumothorax.
- Bleeding.

Endobronchial ultrasound

- EBUS-TBNA offers a minimally invasive approach to biopsying mediastinal nodal tissue and other pathology adjacent to the airways.
- EBUS-TBNA may not be ideal for diagnosis of lymphoma given the small amounts of tissue obtained.
- Current EBUS specialty bronchoscopes have a 7.5MHz convex array linear US probe. When the linear probe is coupled with the bronchial wall, 5cm penetration depth US images are obtained.
- EBUS scopes have a 2mm biopsy channel, with 35° oblique view.
- TBNA utilizes a 22G needle (Wang) suction aspirator system passed through the scope working channel.
- When aspirating a lymph node, lymphocytes must appear in the specimen for the sample to be considered diagnostic.
- Diagnosis of lymphoma may be problematic, given the small-gauge needle and need for a large tissue sample.
- Complications are the same as for flexible bronchoscopy TBNA.

Outcomes

EBUS-TBNA offers 89–99% sensitivity and diagnostic accuracy 85–99%, although a negative predictive value around 90%.

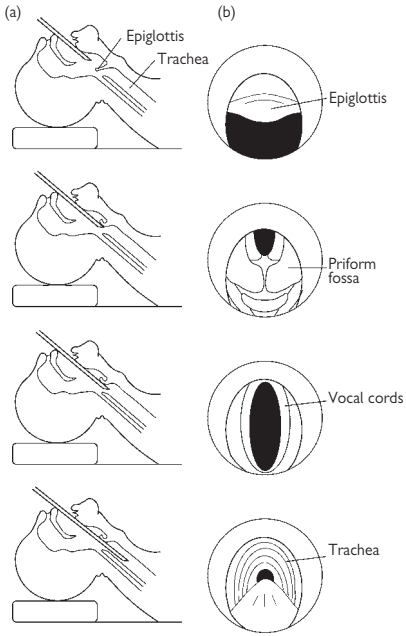


Fig. 13.3 Rigid bronchoscopy. (a) Position of scope. (b) Views as scope enters airway.

Esophagoscopy

Key facts

- Used for investigation of hematemesis, foreign bodies, esophageal strictures, dysphagia, non-cardiac chest pain, symptoms of gastro-esophageal reflux disease, recurrent aspiration pneumonia, odynophagia, abnormal esophageal shadow on chest radiograph.
- May be diagnostic, identifying esophagitis and malignancy, achalasia, webs, diverticula, rings, strictures.
- May be therapeutic, dilating strictures, arresting sites of bleeding, removing foreign bodies, allowing dilation and relief of obstruction.
- There are two main types:
  - Rigid: requires general anesthetic.
  - Flexible: may be performed under sedation as an outpatient procedure.

Flexible esophagoscopy under sedation

Patient must be starved for 6–8h prior to procedure to ensure complete gastric emptying.

- Patient placed in left lateral decubitus position or supine if under general anesthesia.
- Scope is passed by direct vision along tongue to oropharynx. The epiglottis is identified and the scope passed posterior to it, between the piriform recesses into the upper esophagus
- Alternatively the scope can be passed under indirect vision. The scope is placed into the oropharynx, the tongue depressed, and the patient asked to swallow while the scope is slowly advanced. A 'give' is felt when it enters the upper esophagus.
- The esophagus, stomach, and proximal duodenum can be visualized using the flexible scope.
- Scopes may have a double lumen, useful in cases of bleeding.
- Ensure continued visualization upon withdrawing the scope.
- Biopsies and brushings may be taken along with photographs and video recording of any abnormality.

Rigid esophagoscopy under general anesthesia

Patient undergoes general anesthetic and ET intubation. Ensure ET tube is securely fastened.

- Patient supine with neck extended.
- Scope is passed under direct vision into oropharynx.
- It is passed behind the ET tube, between the piriform recesses into the upper esophagus.
- The scope visualizes the esophagus only but allows larger instrumentation than the flexible esophagoscope.
- It also allows the passage of esophageal stents under direct vision.

Normal endoscopic distances from the incisors

- Aortic compression at 25cm (see Fig. 13.4).
- Squamocolumnar junction 40cm:
  - Whitish smooth mucosa above the Z line.
  - Reddish mucosa at and below the Z line.

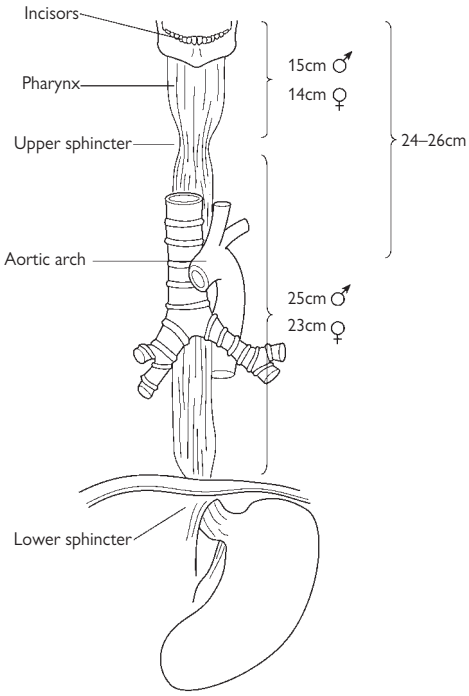


Fig. 13.4 Technique of esophagoscopy.

Mediastinoscopy

Key facts

- Mediastinoscopy is the gold standard for staging mediastinal paratracheal lymph nodes in the setting of lung cancer.
- It is also key in evaluating other pathology of the middle mediastinum.
- Mediastinotomy or Chamberlain procedure is useful for biopsying large anterior mediastinal masses.
- Aortic-pulmonary (AP) artery window lymph nodes can be evaluated by mediastinotomy or now more commonly by left VATS.
- Thorough understanding of the mediastinal anatomy is critical to avoid catastrophic injury (📖 pp40–41).

Indications

- Pre-lung resection staging for primary lung cancer.
- Biopsy of mediastinal masses and lymph nodes for histological tissue diagnosis (e.g., sarcoid, lymphoma, mediastinal tumours, tuberculosis).

Technique of cervical mediastinoscopy

Originally described in 1954.[1] Allows for biopsy of lymph node stations 7 (subcarinal), 4, 2, and 1 (para- and pretracheal regions), as well as some superior mediastinal masses (see Fig. 13.5). Extended mediastinoscopy may access station 10 lymph nodes, though this carries additional operative risk. Although all mediastinoscopes allow the surgeon to look through the eyepiece, most cervical mediastinoscopy can be performed with a Hopkin's telescope, and video display of the procedure on a high-definition monitor (video cervical mediastinoscopy).

- Position patient supine with a transverse shoulder roll to fully extend the neck. The head is placed on a support ring, and should be flush with the superior most edge of the table.
- The entire neck and anterior chest is prepped, with the ability to perform a median sternotomy if necessary. Surgeon stands behind head.
- A pulse oximetry probe is placed on the right hand. If excessive levering causes occlusion of the innominate artery, the pulse oximeter will lose its tracing, alerting the surgeon to relax the mediastinoscope.
- A 1.5cm transverse cervical incision is made one fingerbreadth above the sternal notch, preferably in a pre-existing skin crease.
- The platysma is divided transversely, and the median raphe of the strap muscles are divided longitudinally and the strap muscles are retracted laterally, exposing the pretracheal fascia.
- The pretracheal fascia is divided sharply with Metzenbaum scissors, elevated, and the plane along the trachea is developed with the middle finger and extended down to the carina, with left-to-right blunt dissection of the space. In the process, the locations of the carina, innominate artery, and arch of the aorta are palpated and located.
- Pass the mediastinoscope along the anterior surface of the trachea beneath the pretracheal fascia and beneath the innominate artery which can be palpated.
- Care must be taken to clearly have the tracheal rings in view. Bluntly dissect with the suction cautery before advancing the scope. Advance the scope with two hands, to avoid sudden jerky movements. If disoriented, always return to the tracheal rings to re-establish anatomy.

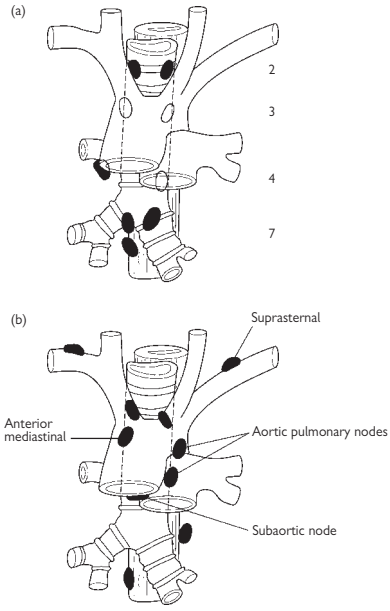


Fig. 13.5 (a) Standard lymph node stations accessible by standard cervical mediastinoscopy (2, 3, 4, and 7). (b) Lymph nodes not readily accessible by this approach.

- Prior to biopsy, identify the carina, then the right mainstem, followed by the left mainstem.
- In order of biopsy:
  - Station 2R, 4R, 4L, 2L if present, and 7.
  - Station 7 is biopsied last as there is frequently nuisance bleeding when biopsying this station. This bleeding nearly always stops with simple packing and time. Avoid cautery in this area because of the proximity of the right main PA.
- Important anatomical points to avoid injury:
  - 4R is near the azygos vein, and pleura of the lung.
  - 2R may be near the innominate artery, pleura of the lung, and the SVC.
  - 4L is near the recurrent laryngeal nerve and arch of the aorta.
  - 7 is near the right main PA and esophagus.
- Tissue is biopsied by grasping it with biopsy forceps, and gently twisting with supination and pronation and gentle traction. Never forcefully pull on tissue, as this may inadvertently avulse an adherent vessel.
- If in doubt whether a structure is vascular or solid, attempt to aspirate with a spinal needle before biopsy.

Reference

1. Harken DE *et al.* A simple cervicomediastinal exploration for tissue diagnosis of intrathoracic disease. *New Engl J Med* 1954; **251:**1041.

Medianstinotomy

Key facts

- Also known as the ***Chamberlain procedure***, left anterior mediastinotomy was first described in 1966.
- This procedure may be performed in conjunction with a cervical mediastinoscopy allowing digital palpation and assessment of the AP lymph nodes. Also very useful for the incisional biopsy of large anterior mediastinal masses.

Procedure

- Position the patient supine with the left side elevated at 30°.
- Make a 4cm left transverse incision along the second intercostal space starting just lateral to the left sternal border.
- Sharp dissection through the intercostal muscle avoiding the internal thoracic vessels.
- Avoid entering the pleural space by blunt dissection of the pleura down to the AP window.
- Biopsy may be direct or via a mediastinoscope passed in through this incision.
- The AP window can also be biopsied by left VATS. Port placement and positioning are described elsewhere (📖 pp686–687).

Right anterior mediastinoscopy

Position as for Chamberlain procedure, but with the right side raised. Allows assessment of the mediastinum, right hilum, and SVC.

- As for the Chamberlain procedure but starting just lateral to the right sternal border and passing through the right second interspace.
- The right mediastinum, SVC, and right hilum can also be explored by right VATS. Port placement and positioning are described elsewhere (📖 pp686–687).

Outcomes

Cervical mediastinoscopy is very effective in staging the mediastinum and generally safe:

- Cervical mediastinoscopy in 947 patients with NSCLC. 2.1% false negative rate for accessible lymph nodes. 0.6% morbidity , 0.2% mortality.[1]

Complications of mediastinoscopy and mediastinotomy

Complications can be catastrophic.

- Tracheal injury, hemorrhage, wound infection, pneumothorax, recurrent laryngeal nerve injury (most common on left), esophageal injury (very rare).
- Azygos vein injury:
  - Usually occurs with biopsy of station 4R.
  - Often avoided by sampling indeterminate lymph nodes with a spinal needle prior to biopsy.
  - After the injury, packing the mediastinum with sponges for 10min often stops the bleeding as the azygos vein is part of a low-pressure system.

- Innominate artery, pulmonary artery, aortic or SVC injury:
  - Will not stop with packing, however, packing allows time to proceed to definitive therapy.
  - The innominate artery is often injured by excessive levering of the cervical mediastinoscope.
  - The PA and aorta may be injured during biopsy of stations 7 and 4L.
  - The SVC may be injured during biopsy of station 2R.
  - After packing, a median sternotomy is performed, providing adequate access of the earlier listed vascular structures. Repair may require establishment of cardiopulmonary bypass.
- Esophageal injury:
  - Esophageal injury may occur during biopsy of station 7. It is often identified a few days after the procedure and may present with dysphagia, odynophagia, fever and/or early sepsis.
  - Diagnosed with endoscopic evaluation and thin barium esophagram. Chest CT may identify mediastinal emphysema and an abscess.
  - The esophagus can be endoscopically stented, and the mediastinum drained via a left lateral collar incision.
- Inadvertant biopsy of the lung:
  - Often occurs on the right.

If air is leak is identified, a right chest tube may be necessary, with a Penrose drain placed in the mediastinum via the cervical incision.

Reference

1. Hammoud Z *et al.* The current role of mediastinoscopy in the evaluation of thoracic disease. *J Thorac Cardiovasc Surg* 1999; **118**:894–9.

Tracheostomy

Key facts

A tracheostomy is normally performed electively, if the patient proves difficult to wean from ventilation after 7–10 days. A tracheostomy may be performed as a formal open surgical procedure, or as a percutaneous tracheostomy using Seldinger technique and serial stoma dilation (📖 p166). A tracheostomy can be performed in the OR, or in the ICU with a skilled support team.

Indications

- Protection of the airway, i.e., from aspiratory secretions, or disrupted anatomy from trauma.
- Prolonged ventilatory support.
- Tracheobronchial toilet (particularly in pneumonectomy patients).

Contraindications

- Local sepsis and hemodynamic instability are relative contraindications.

Technique of open tracheostomy

- Most patients are able to undergo an open tracheostomy despite any body habitus. If the innominate artery is encountered, a strap muscle can be rotated to buttress the innominate artery from the tracheal stoma, and prevent the formation of a TIF.
- The patient is preoxygenated.
- With the patient positioned with the neck straight and extended, and the cuff on the ET tube deflated, the midline is carefully identified: this is critical to prevent malpositioning of the tracheostomy tube in the soft tissues adjacent to the trachea.
- The isthmus of the thyroid overlies the second to fourth tracheal rings: an approach higher than this avoids the isthmus but is associated with greater incidence of tracheal stenosis.
- Most practitioners elect to enter the trachea below the second or third ring.
- The patient's neck is extended, and a shoulder roll is used to elevate the shoulders.
- A 1–2cm incision is made about one fingerbreadth above the sternal notch.
- The platysma muscle is divided transversely and the median raphae of the strap muscles is identified and divided longitudinally.
- The strap muscles are distracted laterally, exposing the pretracheal fascia.
- If the isthmus of the thyroid is encountered, this can be divided with cautery, or suture ligated between clamps.
- The appropriate sized tracheostomy tube is selected (generally #8 or # 6 Shiley).
- The location of the stoma is chosen, and on either side of this area, traction stitches are inserted with 2/0 Prolene. This allows for elevation of the trachea into the surgical view.

- There are several techniques for creating a stoma; the author prefers removing a square segment of cartilaginous ring after cutting with a 15 blade.
- The FiO_2 is decreased at this point, to prevent blaze during cautery for hemostasis.
- After removal of the segment, the stoma can be dilated with a specialized tracheostomy spreader, or snap.
- The cuff on the ET tube is deflated, and the ET tube is backed out under direct vision from the surgical field.
- After the tip of the ET tube has passed from view of the stoma, the lubricated tracheostomy tube is inserted, the obturator removed, and the cuff inflated, and ventilator tubing connected.
- The airway can be suctioned from the field.
- 2/0 Prolene sutures are used to suture the phalanges of the appliance directly to the skin.

Inadvertent extubation and loss of airway control

- In tracheostomies over 7 days old a tract makes reinsertion of the tracheostomy tube relatively easy.
- Prior to the 7 days (including during the tracheostomy procedure itself), the tracheal stoma should not be blindly recannulated, as the tracheostomy tube has a high incidence of placement into a false passage, and consequently development of hypoxic arrest.
- The patient should be bag ventilated, and then re-intubated orally with an ET tube. If this fails, the track can be re-explored under direct vision, or a cricothyroidotomy should be performed.

Tracheoinnominate fistula (TIF)

Catastrophic late bleeding suggests formation of a TIF. It is often preceded by a small herald or sentinel bleed, and may be associated with a pulsating tracheostomy. If a sentinel bleed is suspected, the patient can be studied via a CT angiogram, or a fiberoptic and/or rigid bronchoscopy in the OR with instruments on the ready.

- To gain emergency control, the cuff of the tracheostomy tube can be overinflated, and if this fails then compress the innominate artery against the sternum by removing the tracheostomy and placing a finger inside the stoma. ET intubation must be carried out with a cuffed tube to achieve airway control.
- The patent is brought to the OR, and a median sternotomy is performed.
- The upper part of the pericardium is opened to expose the ascending aorta and origin of the innominate artery.
- The artery is controlled proximally and distally while the left innominate vein is retracted caudal.
- The innominate artery is divided proximally and distally with a running 4/0 Prolene and the tracheal defect is closed primarily with 4/0 Vicryl.
- Sternocleidomastoid, omentum, or other rotational flap is used to buttress the tracheal injury and prevent refistulization.
- Flow can be re-established to the right carotid artery with an interposition graft.

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